Helper-Sensitive Energy Readout and Domain-Correct Error Budgets for Thermodynamic-First QAOA on the SK Model

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Abstract

We study the fixed-calculation interfaces in the manuscript claiming that QAOA reaches the Sherrington–Kirkpatrick ground-state energy when system size tends to infinity before circuit depth. Our main local result replaces a square-root helper-label loss in its final classical-synthesis energy readout by a linear loss. The additional structure is not a generic continuity estimate: insertion vectors have conditional norm one, and an excluded classical target has no helper-supported child first-chaos row. An explicit leakage version quantifies failure of the latter condition. Finite Fock examples show that linear order is sharp under the abstract hypotheses, whereas omission of the row condition restores square-root loss. We also give domain-correct finite-word error budgets allowing loss of weighted powers, and physical masked-Gram certificates which retain omitted-vector errors. These deductions do not prove the source's global classical optimization, helper synthesis, Gaussian-seed removal, or thermodynamic identification independently, and give neither a circuit-depth rate nor a joint size–depth limit. Contribution assessment: C3/5, a source-specific linear helper-loss refinement with sharpness and leakage controls, not a new global SK optimization theorem.

1 Normalization and the Question Actually Addressed

For independent standard real Gaussians J_{ij}, put

$$C_n = n^{-1/2} \sum_{1\le i<j\le n} J_{ij} Z_i Z_j, \qquad B_n = \sum_{i=1}^{n} X_i, \qquad h_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j.$$

The source maximizes this positive-sign Hamiltonian. Its constant is $P_* = \lim_n n^{-1}\mathbb{E}\max_\sigma h_n(\sigma)$, with covariance $\mathbb{E}h_n(\sigma)h_n(\tau) = nR(\sigma,\tau)^2/2 - 1/2$. Adding an independent configuration-constant Gaussian of variance $1/2$ changes that covariance to $nR^2/2$ and does not change the expected maximum. This convention must not be exchanged silently with a negative-energy or ordered-pair convention.

For a fixed depth p and deterministic real angles,

$$U_{n,p} = e^{-i\beta_p B_n} e^{-i\gamma_p C_n} \cdots e^{-i\beta_1 B_n} e^{-i\gamma_1 C_n}, \qquad v_p(\boldsymbol{\gamma},\boldsymbol{\beta}) = \lim_{n\to\infty} \frac{1}{n}\mathbb{E}\left\langle +^{\otimes n}, U_{n,p}^* C_n U_{n,p} +^{\otimes n}\right\rangle.$$

Existence and the tree identification of this fixed-word limit are external inputs in the source. Its assertion is

$$\lim_{p\to\infty} \sup_{\boldsymbol{\gamma},\boldsymbol{\beta}} v_p(\boldsymbol{\gamma},\boldsymbol{\beta}) = P_*.$$

An existence proof of this statement selects a finite word for each accuracy *before* taking $n \to \infty$. It does not establish a size-dependent depth schedule, an efficient angle-finding algorithm, strong classical simulation of sampled bitstrings, or a gap-free adiabatic theorem. Our improvement concerns one error term after a classical target and its helper labels have already been selected.

2 The Rooted Hilbert Space and Its Physical Energy

Let (S, μ) be a probability seed space with a measure-preserving involution $s \mapsto -s$. The local space is $\mathfrak{k} = \mathbb{C}^2 \otimes L^2(S, \mu)$, with vacuum $\omega = |+\rangle \otimes 1$. The source recursively attaches symmetric Fock spaces over centered child spaces and embeds shorter trees by empty descendants. We use its completed pointed space $(\mathcal{H}, \Omega)$. All vectors used below can first be taken at finite height, then limits taken in the Hilbert norm. The child and root copies of this pointed space are identified by the recursive embedding.

On $\mathcal{H}_0 = \Omega^\perp$ define the root-creation isometry

$$Lv = \omega \otimes a^*(v)\Omega_{\mathrm{F}}, \qquad L\Omega = 0. \tag{2.1}$$

Thus $\|L\| = 1$, $L^*L = I - |\Omega\rangle\langle\Omega|$, and L^* extracts the one-child, root-vacuum coefficient. The readout is

$$\mathcal{E}(v) = \mathrm{Re}\,\langle v, Lv\rangle\,, \qquad |\mathcal{E}(v) - \mathcal{E}(w)| \le (\|v\| + \|w\|)\,\|v - w\|\,. \tag{2.2}$$

Inner products are conjugate linear in the first argument. The energy identity with the limiting rescaled tree edge observable, rather than just this abstract quadratic form, is a source input. In particular its finite-degree proof uses a uniform expected occupation bound in addition to strong convergence.

Joint spin/seed parity fixes Ω and sends Z to $-Z$. We call a vector odd when it changes sign under that parity; odd vectors are centered. An even seed set H defines the root multiplication projection $P = 1_H$; write $Q = I - P$ and $p_H = \mu(H)$. These projections preserve parity but need not preserve the centered subspace for arbitrary, nonodd vectors. Below we stay in the odd subspace. The same symbol P on L^*w denotes the corresponding *child-root* mask under the recursive identification.

Lemma 2.1 (Conditional insertion normalization)**.** *If $u = \int^{\oplus} u(s)\,d\mu(s)$ is a seed-decomposable unitary, preserves joint parity, and $v = u^*Zu\Omega$, then v is odd and $\|v(s)\| = 1$ almost everywhere. Consequently $\|Pv\| = \sqrt{p_H}$ and $\|Qv\| = \sqrt{1 - p_H}$. Moreover, for centered v, $\|(Lv)(s)\| = \|v\|$ almost everywhere.*

Proof. The vacuum has conditional norm one, Z and $u(s)$ are unitary, and joint parity anticommutes with Z. The last assertion follows directly from (2.1): the root seed function is identically one. Integrating the conditional squared norms proves the two mask formulas. No differentiation of seed masks, nor independence between masks and other root seed functions, is used. □

3 A Linear Helper-Loss Readout

Theorem 3.1 (Readout with a measured first-chaos leak)**.** *Let $0 \le p_H < 1$, and let v, w_0 be centered odd vectors such that $\|v(s)\| = 1$ almost everywhere, $Pw_0 = 0$, and $\|w_0(s)\| = 1$ almost everywhere on $S \setminus H$. Set*

$$\eta = \|Qv - w_0\|\,, \qquad d = \|PL^*w_0\|\,.$$

Then

$$|\mathcal{E}(v) - \mathcal{E}(w_0)| \le \big(2\sqrt{1 - p_H} + \sqrt{p_H}\big)\eta + p_H + \sqrt{p_H}\,d. \tag{3.1}$$

In particular, if $PL^*w_0 = 0$, the bound is at most $\sqrt{5}\,\eta + p_H$.

Proof. Write $w = Qv$, $h = Pv$, and $e = w - w_0$. Exact expansion gives

$$\mathcal{E}(v) - \mathcal{E}(w_0) = \mathcal{E}(w) - \mathcal{E}(w_0) + \operatorname{Re}\langle h, Lv\rangle + \operatorname{Re}\langle w, Lh\rangle . \tag{3.2}$$

Since $\|w\| = \|w_0\| = \sqrt{1-p_H}$, (2.2) bounds the first term by $2\sqrt{1-p_H}\eta$. Conditional normalization, not merely global unit norm, gives

$$|\langle h, Lv\rangle| \le \int_H \|v(s)\|\,\|(Lv)(s)\|\,d\mu(s) = p_H.$$

Finally $|\langle w_0, Lh\rangle| = |\langle PL^*w_0, h\rangle| \le d\sqrt{p_H}$, whereas $|\langle e, Lh\rangle| \le \eta\sqrt{p_H}$. Their sum proves (3.1); Cauchy–Schwarz gives $2\sqrt{1-p_H} + \sqrt{p_H} \le \sqrt{5}$. □

3.1 Why the Source's Exclusions Give the Row Condition

The classical subspace has all physical spins in $|+\rangle$ and real seed/Gaussian coordinates. For real centered classical child vectors f_j, the commuting positions $q(f_j)$ have joint Gaussian covariance $\langle f_i, f_j\rangle$. Their covariance is allowed to be singular.

Lemma 3.2 (Excluded Gaussian directions)**.** *Suppose a finite classical calculation at its root uses, for almost every root seed, a square-integrable function of finitely many commuting positions $q(f_1), \dots, q(f_m)$, where the real classical directions satisfy $Pf_j = 0$. Allow bounded seed-dependent coefficients and a finite partition of the root seed space into formulas of this type. If w_0 is its root-$|+\rangle$ part plus any root-$|-\rangle$ part, then $PL^*w_0 = 0$.*

Proof. Let V be the real span of the f_j at a fixed seed formula. Choose an orthonormal basis of V, omitting redundant directions. Gaussian factorization of the classical Fock space over $V \oplus V^\perp$ shows that a function of these positions has first chaos in V: its pairing with any one-particle vector in $V^\perp$ is zero by independence and centering. This is the elementary orthogonal-chaos projection fact, not a covariance-inversion formula. The root-$|-\rangle$ part is killed by L^*. Averaging the root seed in the remaining coefficient gives a Bochner integral in the closed subspace $\ker P$, because every first-chaos coefficient belongs to that subspace. Finite seed partitions cause no change. The projection bound by the original vector norm justifies that integral. □

In the source's colored, ancestor-excluded classical target, each aggregation direction excludes the entire helper set. Different branches can share directions and masks; independence between those branches is not required in Lemma 3.2. At fixed finite calculation all root formulas use finite direction lists, so the lemma applies separately on the finite root-type partition. It would *not* apply to an arbitrary quantum target with helper-supported child directions.

More explicitly the source chooses a real smooth angle θ, zero on helpers, and writes

$$v_{\mathrm{cl}} = |+\rangle m + |-\rangle \cos(2\theta), \qquad m = \sin(2\theta), \qquad w_0 = Qv_{\mathrm{cl}}. \tag{3.3}$$

Here m and $\cos(2\theta)$ are classical descendant vectors, with the identity $\sin^2(2\theta) + \cos^2(2\theta) = 1$ in their commuting functional calculus. Thus w_0 has conditional norm one off helpers. The helper part of v_{cl} is $1_H|-\rangle$ with empty descendants. Its left pairing with L is zero because its root spin is $|-\rangle$; its right pairing with the classical target is zero because it creates a child spin excitation. The same spin orthogonality removes the cosine terms. Hence $\mathcal{E}(w_0) = \mathcal{E}(v_{\mathrm{cl}}) = \mathcal{E}(m)$.

Corollary 3.3 (Source-specific synthesis readout)**.** *Assume the source's fixed classical target (3.3), its helper exclusions, and its seed-decomposable synthesized insertion vector v with off-helper error at most η. Then*

$$|\mathcal{E}(v) - \mathcal{E}(m)| \le p_H + \big(2\sqrt{1-p_H} + \sqrt{p_H}\big)\eta. \tag{3.4}$$

This improves the source's final $2\eta + 2\sqrt{p_H}$ readout in its helper-probability order; it does not improve every coefficient of η. For example $p_H = 10^{-4}$ and $\eta = 10^{-4}$ give a new bound below $301 \cdot 10^{-6}$, versus $202/10^4$ from the source estimate. This is an exact bound comparison, not a circuit experiment. The earlier classical coloring/exclusion approximation still costs $O(\sqrt{\alpha + p_H})$, and helper contractions can have constants depending on inverse helper probabilities. Those costs remain.

4 Sharpness and Missing-Hypothesis Examples

The following are finite-height vectors in the very space above. They demonstrate sharpness of the local hypotheses, not reachability by a particular ordinary QAOA word. Choose a nontrivial even mask of probability $p \in (0, 1)$ and put $r = |-\rangle$ with empty descendants.

Proposition 4.1 (Linear order cannot be improved)**.** *Let $w_0 = Qr$, $e_0 = Qr/\sqrt{1-p}$ and $h = 1_H|+\rangle \otimes a^*(e_0)\Omega_{\mathrm{F}}$, and set $v = w_0 + h$. These vectors satisfy Theorem 3.1 with $\eta = d = 0$ and*

$$\mathcal{E}(v) - \mathcal{E}(w_0) = p\sqrt{1-p}. \tag{4.1}$$

Consequently no universal $o(p)$ replacement is possible.

Proof. Both conditional norms are one on their respective supports, and the vectors are odd. The root-minus spin gives $L^* w_0 = 0$. The only nonzero energy pairing is $\langle h, Lw_0\rangle = p\,\langle e_0, w_0\rangle = p\sqrt{1-p}$. In particular the pairing involving Lh vanishes: the child of h has empty descendants, whereas h itself has one child. The source's embeddings make these orthogonal. □

Proposition 4.2 (The first-chaos row is essential)**.** *Put $e_H = Pr/\sqrt{p}$, $w_0 = 1_{S\setminus H}|+\rangle \otimes a^*(e_H)\Omega_{\mathrm{F}}$, $h = Pr$, and $v = w_0 + h$. Conditional normalization and $\eta = 0$ still hold, but*

$$PL^* w_0 = (1-p)e_H, \quad d = 1 - p, \quad \mathcal{E}(v) - \mathcal{E}(w_0) = \sqrt{p}(1-p). \tag{4.2}$$

Proof. Root-vacuum averaging contributes $\mu(S \setminus H) = 1 - p$ to $L^* w_0$. The sole surviving pairing is $\langle w_0, Lh\rangle = (1-p)\,\langle e_H, Pr\rangle$. All other pairings vanish by the root-spin or descendant-number orthogonality used above. The target here has a helper-supported child spin excitation and is not an excluded classical target. □

Finally a global unit vector need not have conditional norm one: $p^{-1/2}Pr$ is globally normalized but has helper mass one for every $p > 0$. A small seed probability alone therefore cannot justify discarding a vector supported there.

5 Finite-Word Budgets with Loss of Weighted Powers

The source uses positive weights $W = 1 + d\Gamma(\Lambda)$, with an additional root seed-moment weight when seed coefficients grow. Let $X_a = \mathrm{Dom}(W^a)$, $\|\xi\|_a = \|W^a \xi\|$, for $a \geq 0$, and $\mathcal{S} = \bigcap_{a \geq 0} X_a$. A smooth bound means that each requested output power a has some input power b and constant $C_{a,b}$ with $\|T\xi\|_a \leq C_{a,b}\,\|\xi\|_b$. It does not mean that X_a is invariant with a bound on that same norm. This distinction is important for repeated field operations.

Proposition 5.1 (Domain-correct telescoping)**.** *Let U_j, V_j preserve $\mathcal{S}$, $1 \leq j \leq N$, and put $U = U_N \cdots U_1$, $V = V_N \cdots V_1$. Suppose, for given nonnegative powers a_j, b_j, a and nonnegative constants ϵ_j, M_j,*

$$\|(U_j - V_j)\xi\|_{a_j} \leq \epsilon_j\,\|\xi\|_{b_j}, \qquad \|V_N \cdots V_{j+1}\zeta\|_a \leq M_j\,\|\zeta\|_{a_j}. \tag{5.1}$$

The second bound includes the empty suffix. If $B_j(\xi) = \|U_{j-1}\cdots U_1\xi\|_{b_j} < \infty$, *then*

$$\|(U-V)\xi\|_a \le \sum_{j=1}^{N} M_j\epsilon_j B_j(\xi). \tag{5.2}$$

The formula extends by graph-norm completion whenever all displayed operators and estimates extend on the stated domains.

Proof. The algebraic identity on $\mathcal{S}$ is

$$U - V = \sum_{j=1}^{N} V_N\cdots V_{j+1}(U_j - V_j)U_{j-1}\cdots U_1.$$

Apply the suffix estimate, then the local difference estimate, and sum. Completeness gives the extension. Every required intermediate power has appeared explicitly; no invariance of one common X_a has been assumed. □

For Hilbert-norm output and genuinely unitary suffixes one can take $a_j = a = 0$, $M_j = 1$. The input requirements b_j and the propagated prefix norms B_j still remain. To bound a prefix from a single initial norm, recursively choose the input power required by the previous factor for the currently requested output power. This finite backward selection can raise the initial power with N.

Corollary 5.2 (Two forward inputs suffice for insertion error)**.** *Assume* U, V *are unitaries commuting with* Q, *as does* Z, *and* $v_U = U^*ZU\Omega$, $\phi = V^*ZV\Omega$. *Put* $\xi_0 = Q\Omega$, $\xi_1 = Q\phi$ *and assume these inputs have all domains needed in Proposition 5.1. Then*

$$\|Q(v_U - \phi)\| \le \sum_j M_j\epsilon_j\big(B_j(\xi_0) + B_j(\xi_1)\big) \quad (a = 0). \tag{5.3}$$

Proof. Multiplying by U and commuting Q gives $UQ(v_U - \phi) = ZU\xi_0 - U\xi_1$. Since $V\xi_1 = ZV\xi_0$, this is $Z(U-V)\xi_0 - (U-V)\xi_1$. Apply (5.2) twice. No approximation to U^* on an uncontrolled high-occupation vector is needed. □

The source already uses the corresponding unmasked two-input unitarity argument in seed simulation. Here the combination with the helper-sensitive readout makes an explicit conditional error ledger; the underlying telescoping and unitarity identities are classical, not separately claimed as new principles.

5.1 A Fixed Budget and Its Quantifiers

For a selected excluded target, define nonnegative numbers $\epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{seed}}, \tau$ bounding respectively its gap below P_*, seed-removal energy loss, and any separately certified finite-size transfer error for the *chosen* ordinary word. If the weighted propagation supplies η and a row certificate supplies d, then the conditional lower bound is

$$P_* - \epsilon_{\mathrm{cl}} - \epsilon_{\mathrm{seed}} - \tau - p_H - \big(2\sqrt{1-p_H} + \sqrt{p_H}\big)\eta - \sqrt{p_H}d. \tag{5.4}$$

When using only the source's limiting theorem, take $\tau = 0$ at its fixed-word thermodynamic limit, not at an unspecified finite n. No rate for $\tau(n)$ is supplied here. As a readout-only allocation, $d = 0$, $p_H \le \varepsilon/2$ and $\eta \le \varepsilon/(2\sqrt{5})$ suffice for error ε. This allocation is made *after* a target is chosen with its

own classical approximation error. It cannot authorize replacing every $O(\varepsilon^2)$ helper choice earlier in the source by $O(\varepsilon)$.

All bounds in (5.1) used for a degree limit must be uniform in degree for this fixed calculation. Constants uniform in one finite approximation are not thereby uniform in the number of roles, channels, nested commutators or selective probes. In particular the seed-removal construction first fixes a small initialization parameter, perturbs finitely many angles away from visibility zeros, chooses finite probe lists, and only then makes the echo parameter small. Its r^{-1} pulses are fixed numbers before the thermodynamic limit. A norm error $O(r)$ in a field direction cannot be divided by r and declared negligible; the source instead cancels the two *exact* echo directions before estimating the residual.

6 Physical Finite Certificates and What They Omit

The following is a supporting way of recording the hypotheses in Theorem 3.1. It is not a replacement for its infinite Fock-space target or for the source's gate calculus. Let $J : \mathbb{C}^m \to \mathcal{H}$ synthesize a finite vector bank, and set $G = J^*J$, $G_H = J^*PJ$. Linear independence is unnecessary. These are the physical, not coefficient-Euclidean, Gram matrices.

Proposition 6.1 (Masked-row and insertion certificates)**.** *Suppose a candidate error e and target row $z = L^*w_0$ satisfy*

$$e = Jc + r, \quad \|r\| \le \sigma, \qquad z = Jb + t, \quad \|t\| \le \tau.$$

Then valid inputs to (3.1) are

$$\eta \le \sqrt{c^*Gc} + \sigma, \qquad d \le \sqrt{b^*G_Hb} + \tau. \tag{6.1}$$

If $r \perp \operatorname{Ran} J$, the first bound improves to $\eta \le \sqrt{c^*Gc + \sigma^2}$. If a self-adjoint $W \ge I$ has $w_0 \in \operatorname{Dom}(W^s)$, and one uses $z_R = L^*1_{[1,R]}(W)w_0$, the discarded row error is at most $R^{-s}\,\|W^sw_0\|$ for $R \ge 1$ and $s > 0$.

Proof. The two quadratic forms are exactly $\|Jc\|^2$ and $\|PJb\|^2$. Apply the triangle inequality and $\|P\| \le 1$; orthogonality gives the first Pythagorean improvement. For the last assertion, the spectral theorem bounds the discarded part of w_0, and L^* is a contraction. No commutation of L with W is assumed. □

A spectral cutoff is not automatically finite rank. Conversely a finite occupation cutoff can leave infinitely many child modes. One must supply both a physical bank approximation and its remainder, or a separate finite-rank approximation with justified tails. Near-singular G is not a problem for evaluating a known c^*Gc, but recovering coefficients from measurements can be ill-conditioned. A numerical near-zero value of b^*G_Hb is not the exact first-chaos exclusion of Lemma 3.2.

6.1 A Small Rational Masked-Gram Example

Take orthonormal e_1, e_2, e_3, let $Pe_1 = e_1$ and $Pe_2 = Pe_3 = 0$, and let J have columns $e_1 + e_2$ and $e_1 - e_2$. Then $G = 2I_2$, $G_H = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The coefficient vector $b = (1, -1)/2$ gives $Jb = e_2$ and $b^*G_Hb = 0$, although neither bank column individually avoids the helper space. An omitted component τe_1 makes the actual row leak exactly $|\tau|$; dropping that residual would be false. The vector $c = (1, 1)/10$ gives $\|Jc\| = 1/5$. Adding an orthogonal error σe_3 gives squared error $1/25 + \sigma^2$, not the coefficient squared norm $1/50 + \sigma^2$. Thus actual physical metric and omitted components are both essential.

7 Source Architecture and Retained Proof Obligations

The source's scientific text is organized into eight modules. The following map records the full route, not an assertion that its global claims have been independently proved in this report.

Module	Mechanism and retained obligation
Introduction	Fixed-depth deterministic-angle SK value; monotonicity by zero-angle padding; expected ground-state upper bound.
Foundations	Symmetric-occupation embeddings, root-shift energy, weighted Weyl bounds, normalized sums, unitary central limit, commuting substitution, exclusion gates and additive cut responses.
Classical optimization	Zero-temperature Parisi diffusion and full support input; finite Euler calculation, bounded smoothing, trigonometric approximation and ancestor/color/helper exclusions.
Classical synthesis	Sign averaging, local spin control, helper contractions, centered quadratic kernels, scalar phases, bottom-up replacement in a fixed surrounding circuit.
Seed simulation	Exact ordinary/special packet split, zero-clock selective rotations, actual-direction singular echoes, external-mode Gram induction and a separate root-shift energy computation.
Assembly	One finite ordinary word selected before the SK limit; no simultaneous depth–size estimate.
Positive-temperature alternative	Fixed finite temperature Parisi support and PDE inputs, lagged cavity messages, strong symmetric-power realization, occupation control and clipped bounded targets.
Regular MaxCut	Tree-cone sign change, the $2/\sqrt{D}$ angle scaling, high-girth application and a separate successive random-regular size then degree limit.

7.1 Specific Valid Details That Must Survive Reformulation

The source already keeps the helper partition quantity $\rho = (\sum_\ell p_\ell^2)^{1/2}$, rather than replacing everything by the number of cells. Its quadratic helper response includes subtraction of the scalar contraction and estimates both pair-creation and number-preserving kernels. These are source results, not new contributions here. Likewise the additive cut response retains the scalar second-order term after commuting operator substitution, and the zero-boundary subtraction uses $m(y) - m(0)$ even when $m(0) \neq 0$. A covariance matrix alone neither proves those responses nor identifies the self-adjoint time-group generator.

The source's Fourier substitution argument first chooses the frequency decay needed by the desired output graph norm, and only then chooses a sufficiently high input weight. Its adjoint convergence is proved, not inferred from bounded adjoint jets. Every aggregation in the final colored target excludes *all* helpers as well as current and ancestor types. This entire-helper exclusion is exactly what Lemma 3.2 requires.

The seed simulation distinguishes designated packet tails, with controlled top-number moments, from discarded norm-small remainders, on which no such moments are assumed. The latter are carried by unitarity. Its finite Gram convergence is followed by an explicit root-shift pairing; marginal Gaussian laws alone would not identify energy. In the positive-temperature route the fixed mesh limit keeps temperature and cutoff fixed, and the outgoing coefficient is lagged so that the next Gaussian innovation is conditionally centered. None of these distinctions is removed by the present readout

bound.

7.2 Imported Global Inputs and Nonclaims

We import, without an independent complete validation, the source's classical near-optimum construction, global helper synthesis, seed-removal theorem and fixed-word tree-to-SK identification. Their external Parisi/support/PDE, message-passing and random-regular-graph inputs are also imports. Reading their use in this manuscript is not reading or auditing all the cited proofs. The supplementary positive-temperature route removes dependence on the zero-temperature scalar companion but introduces its own finite-temperature full-support input; it is not assumption-free.

No correction to a source identity is asserted here. The final square-root helper readout is a valid coarse inequality, and we refine it using additional structure already present in its target. Our result does not independently settle the eventual Parisi-optimality conjecture, establish any adiabatic gap, or convert an existential finite construction into efficient synthesis.

8 Original Ram Proofs and the Actual Synthesis

Fresh literal reads of three original Volume 21 chapter bodies [6, 7, 8] support the physical interpretation of this report. The state/domain chapter treats an operator together with its graph domain and transports that domain under coordinates. This motivates retaining every input and output power in (5.1), rather than calling a coefficient matrix a bounded physical operator.

The physical-metric chapter proves $G = J^*J$, the projected weak system, and the Pythagorean decomposition of a full residual into retained and omitted parts. Proposition 6.1 specializes those elementary identities to the helper-supported first-chaos row. The spectral-limits chapter gives a nested finite-section spectral-pollution example and separates strong from norm-resolvent convergence. Its lesson here is precise: finite Fock examples test the local formulas but cannot validate the degree limit, its occupation tails, or the global optimization theorem.

These original arguments are classical Hilbert-space tools expressed in the Ram framework. The distinct increment earned here is the use of conditional insertion normalization and helper-excluded first chaos to obtain (3.4), including the exact leakage and sharpness statements. We do not attribute a new general projection principle, chaos decomposition or telescoping identity to this report.

9 Public Literature and Status as of October 9, 2026

The assigned September 25 manuscript is a public preprint claiming a proof; its presence in a repository is not independent certification. The captured October 8 byte-identity record authenticates the text used here, not the current live branch or correctness of its theorem.

Basso, Farhi, Marwaha, Villalonga and Zhou [2] give the fixed-depth large-girth/tree and SK framework; their arXiv version record and abstract were refreshed here. The abstract formulates eventual Parisi optimality as an optimistic conjecture. We use the source's stated fixed-word theorem as an external input, without a fresh full audit of that external proof.

The April 22, 2026 version of Boulebnane et al. [3] describes a rigorous infinite-size spin–boson representation at fixed depth and numerical high-depth approximations. Sels and Morone [4] describe a semiclassical analysis with a proposed large-depth rate. Neither abstract alone establishes the source's full thermodynamic-first attainment claim.

A newer September 30 preprint by Boulebnane, Khan, Subrahmanya, Herman, Farhi, Villalonga and Shaydulin [5] states exact infinite-size decoupling for computing QAOA energy on mixed dense spin glasses, with high-depth numerical applications. Its abstract explicitly distinguishes energy

evaluation from strong simulation of bitstring sampling. Only the abstract and version metadata were inspected here; no theorem from its 108-page proof is independently audited or used to establish the local readout result.

10 Advantages, Costs, and Further Questions

The new readout uses the true seed law, not a fictitious independent mask derivative. It allows degenerate Gaussian direction Grams and has dimension-free constants at the stated Hilbert-space level. The measured-row version also degrades continuously when exact exclusion is replaced by a certified leak.

Its principal limitation is location: this is the final readout interface, not the earlier classical construction. An improved helper-probability term can be dominated by coloring error, weighted prefix norms, inverse helper probabilities or the long seed-removal word. No running-time or gate-count advantage has been measured. Targets outside the excluded classical subspace can have the square-root loss in Proposition 4.2.

A useful future direction is to propagate explicit constants through one fixed helper construction while preserving all mixed cut responses. Another is an a posteriori first-chaos row computation with certified occupation and mode tails. Neither task can use finite coordinate rank as a substitute for an infinite-dimensional tail estimate. A quantitative size–depth theorem would additionally require uniform fixed-word transfer, occupation and synthesis estimates as the chosen words grow; that problem is not resolved by this report.

11 Reading and Verification Boundaries

All eleven assigned cached text files were read literally in bounded untruncated displays: the main file, eight scientific modules, references and README. This is a complete reading of this source text, not a verification of every imported global argument or of its bibliography. The three original Volume 21 chapter bodies were read in full; their separately included figure PDFs were not viewed. A registered nonlinear-jet chapter was discovered as an alternative lead but its body was not read or used. No whole-private-corpus reading is claimed.

The owned reading ledger records pre-display hashes, actual line intervals, unread complements and matching post-read hashes. The executable local checks evaluate finite rooted Fock compressions, masked physical Grams, and ordered product identities. They test the formulas and negative controls, not QAOA optimization, the limiting Parisi value or the source's infinite constructions. Public literature evidence is confined to the access scope specified above. Applicability L3 is independent of the C3 local contribution assessment.

12 Conclusion

The source contains more information than a global unit-vector continuity estimate uses. Conditional normalization pays exactly a helper probability for one pairing, and full helper exclusion removes the other target row. This yields a linear helper-loss readout with explicit leakage and unavoidable linear-order examples. Retaining weighted domain losses and omitted physical-vector components makes that improvement usable as a conditional certificate, while leaving the source's demanding global and thermodynamic obligations visible.

References

[1] OpenAI. *QAOA attains the SK ground-state energy in the thermodynamic-first limit.* September 25, 2026, assigned public preprint, sections 1–8. Public source directory. Cached scientific text read in full; global claims imported as stated.

[2] J. Basso, E. Farhi, K. Marwaha, B. Villalonga and L. Zhou. *The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington–Kirkpatrick Model.* TQC 2022, LIPIcs 232, article 7. arXiv:2110.14206v3. Abstract and version record inspected.

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